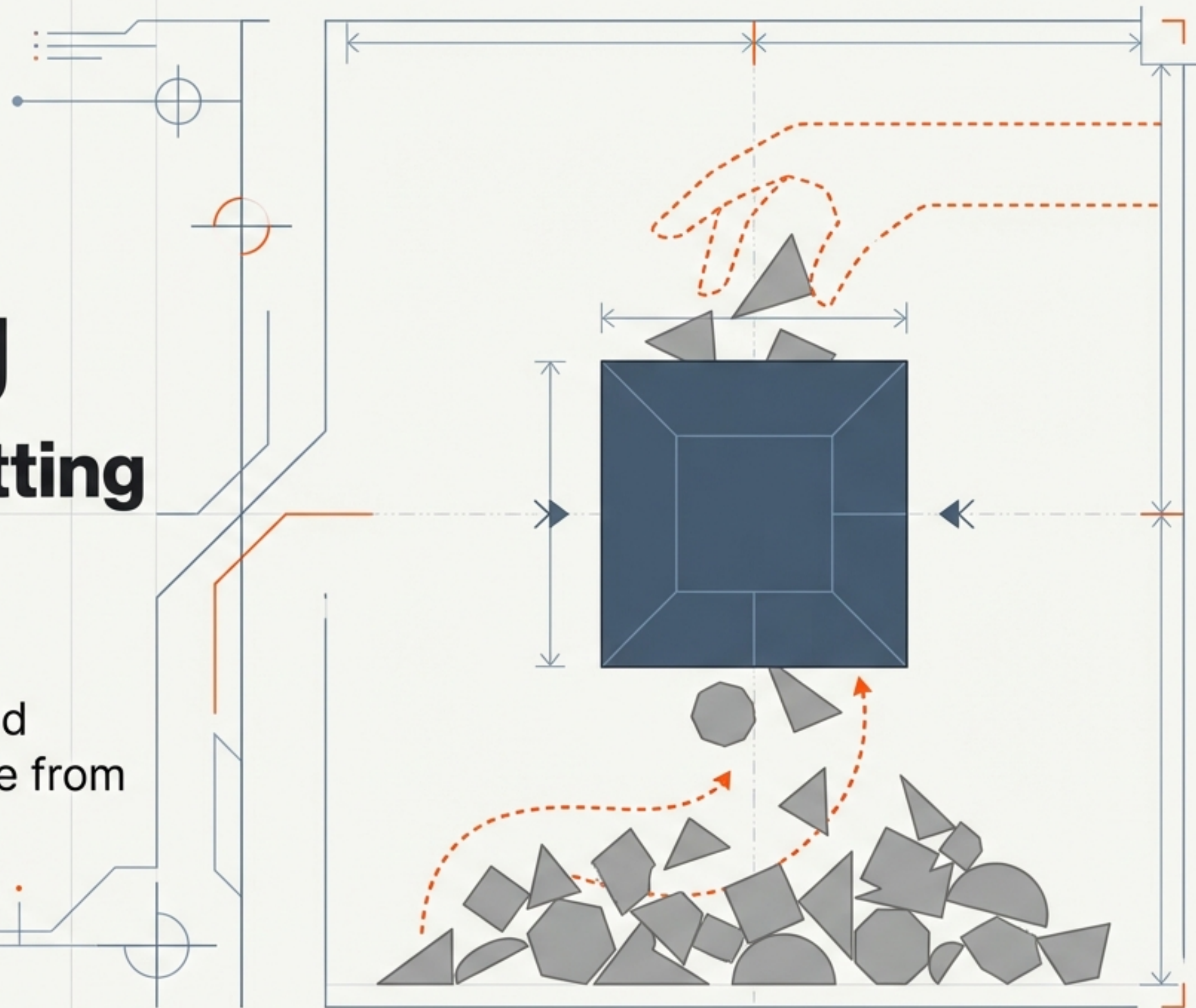


The Art of Feature Engineering

Construction & Splitting

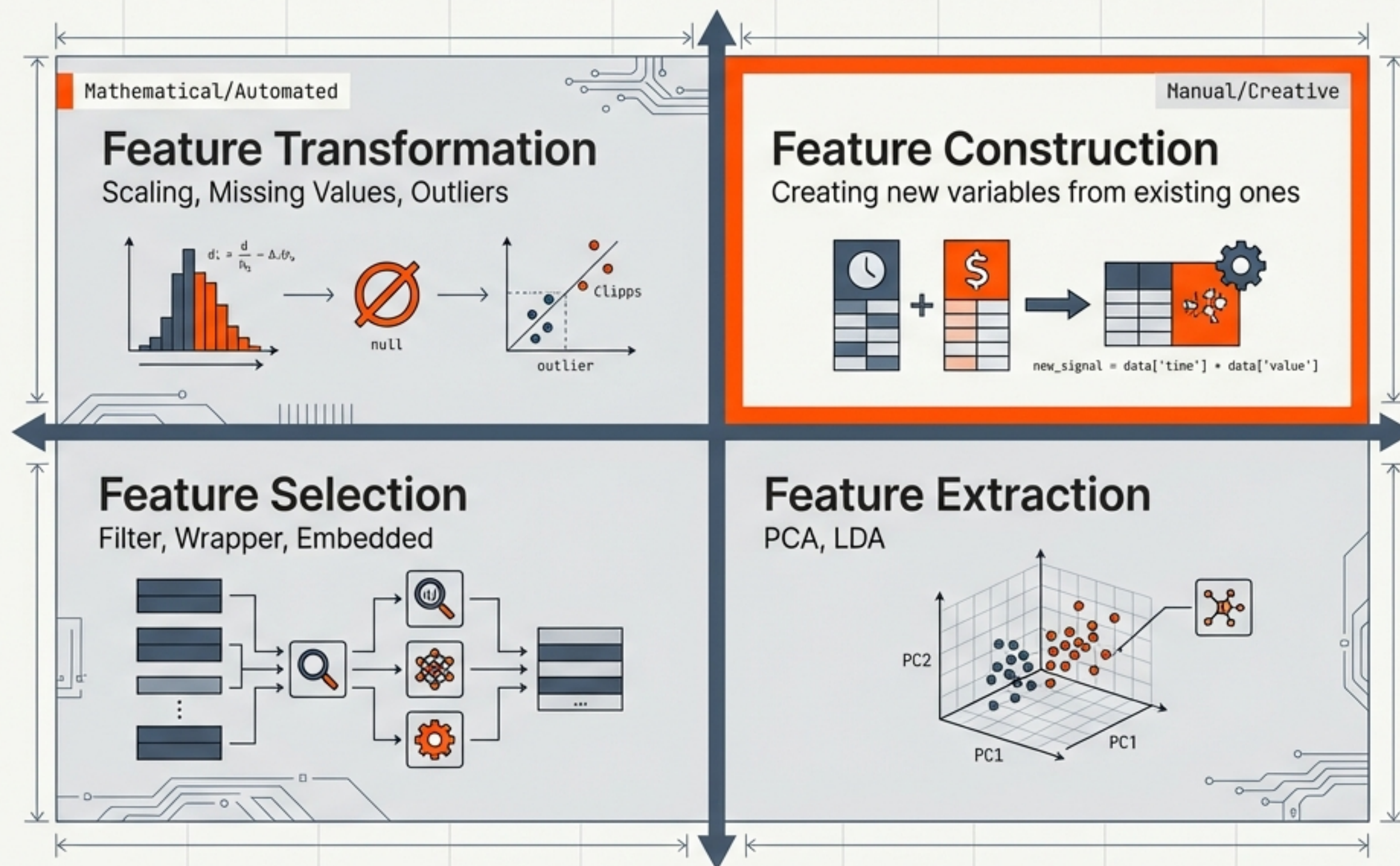
Leveraging domain knowledge and intuition to extract maximum value from raw data.

A Technical Deep Dive

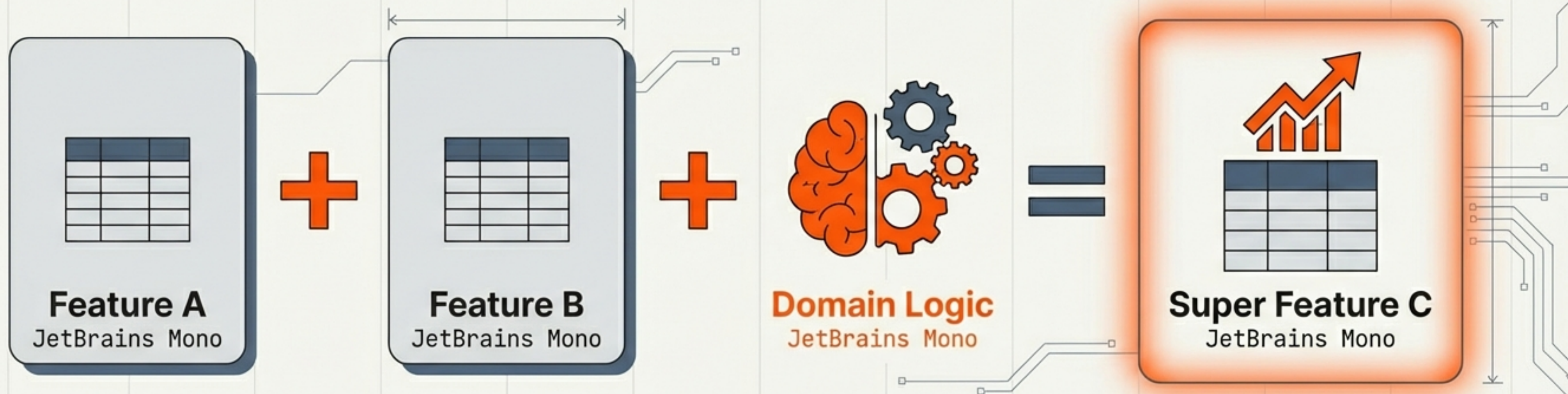


The Landscape of Feature Engineering

While Feature Transformation handles the mathematical hygiene of data (scaling, cleaning), Feature Construction is the creative engine. It is not procedural; it is intuitive. It requires a domain expert to manually hypothesise and build new signals.



Feature Construction: Engineering Intuition



Domain Knowledge

Neue Haas Grotesk Display Pro

Understanding the real-world context (e.g., family dynamics on a sinking ship).

Intuition

Neue Haas Grotesk Display Pro

Spotting relationships that algorithms miss in raw formats.

Experience

Neue Haas Grotesk Display Pro

The ability to recognize patterns across different datasets.

Case Study I: Fragmented Family Data

	PassengerId	SibSp	Parch	Fare
1	1	1	0	27.00
2	2	3	2	30.00
3	3	0	0	112.00
4	...	1	1	172.00
5	...	3	2	325.00

← Problem: Individually, these columns constitute weak signals. A model may struggle to infer 'Group Size' from separate sibling and parent counts.

Hypothesis: Survival chances are likely correlated with the total size of the group a passenger is traveling with, rather than specific biological relationships.

Constructing the 'Family_Size' Feature

SibSp
(Siblings/Spouse)

+ Parch
(Parents/Children)

+ 1 (The Passenger)

= Family_Size

```
df['Family_Size'] = df['SibSp'] + df['Parch'] + 1
```

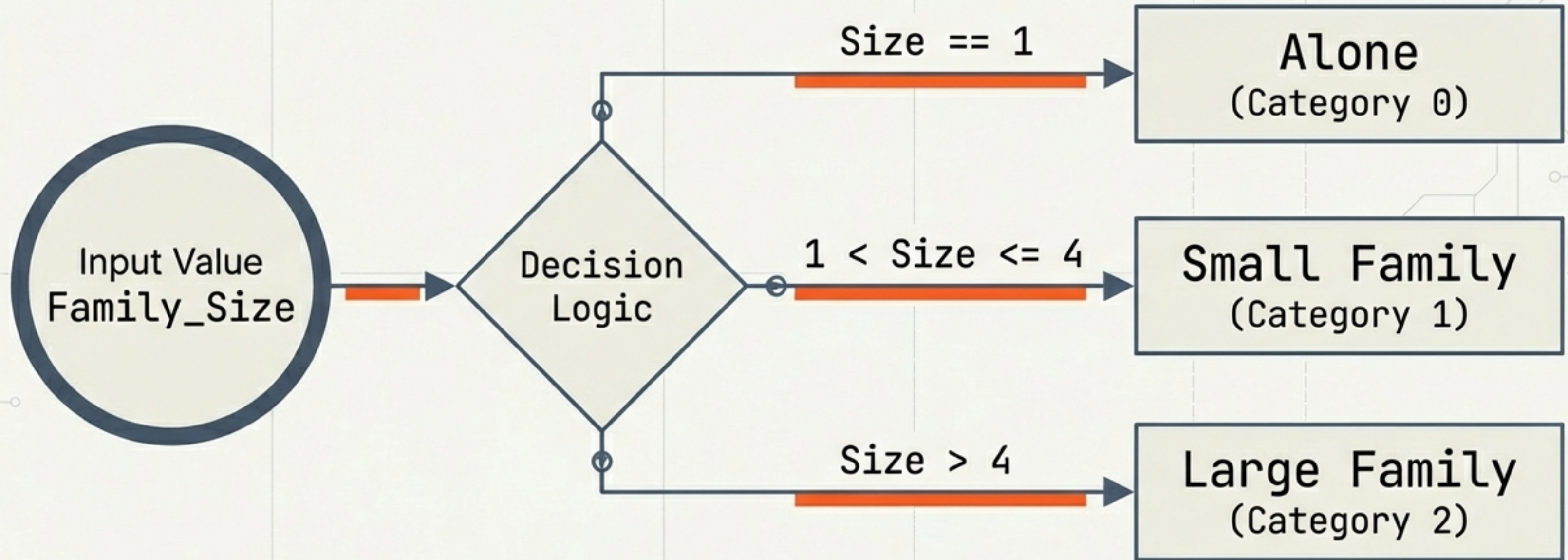
Example:

A passenger with 1 Sibling and 0 Parents.

Calculation: $1 + 0 + 1 = 2$.

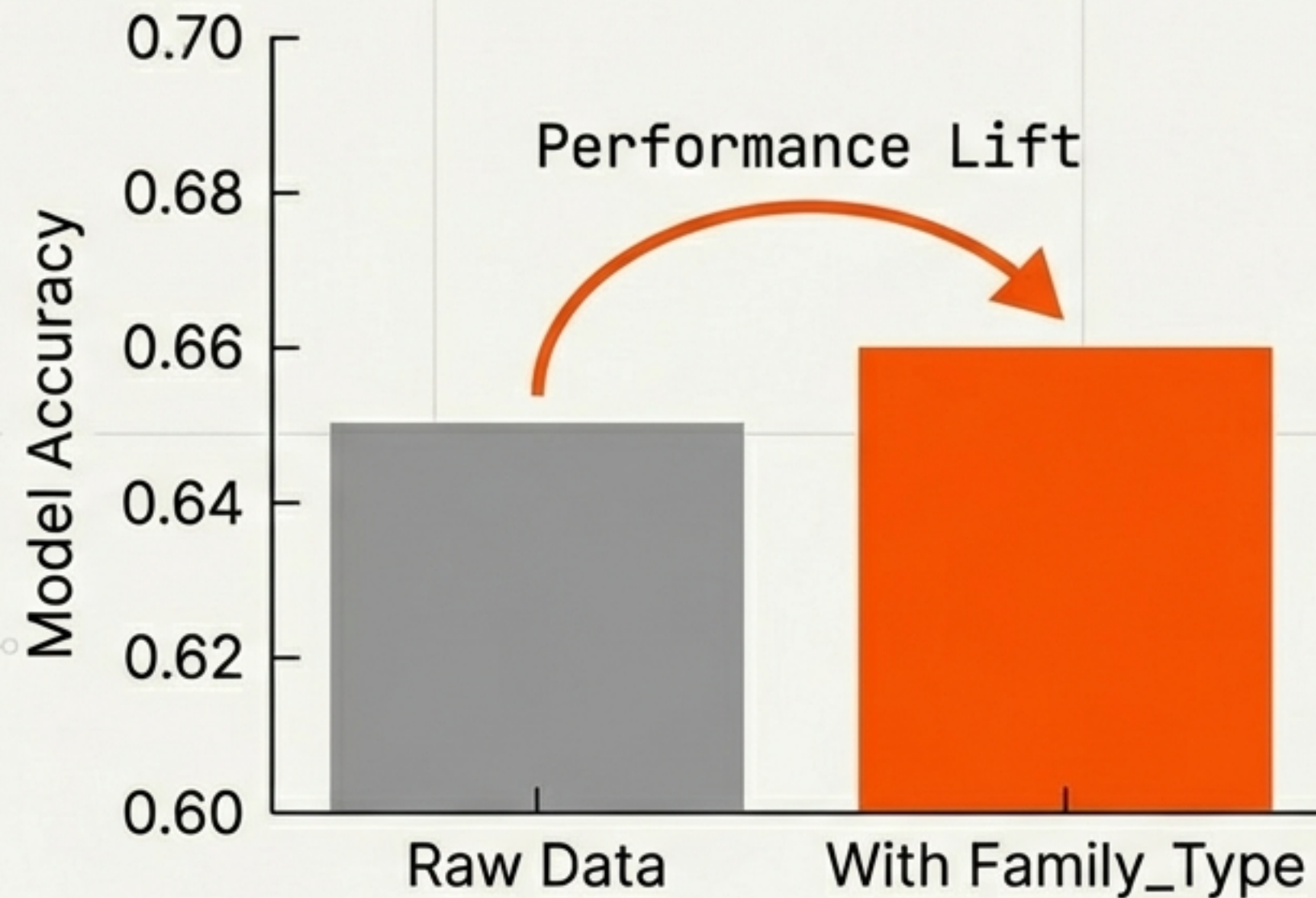
Result: The model now sees a 'Group Unit' of 2, rather than disconnected integers.

Refining Signal through Categorisation



Outcome: We translate a raw continuous integer into semantic categories. This 'binned' feature often carries stronger predictive weight than the raw number.

Measuring Impact: Before and After



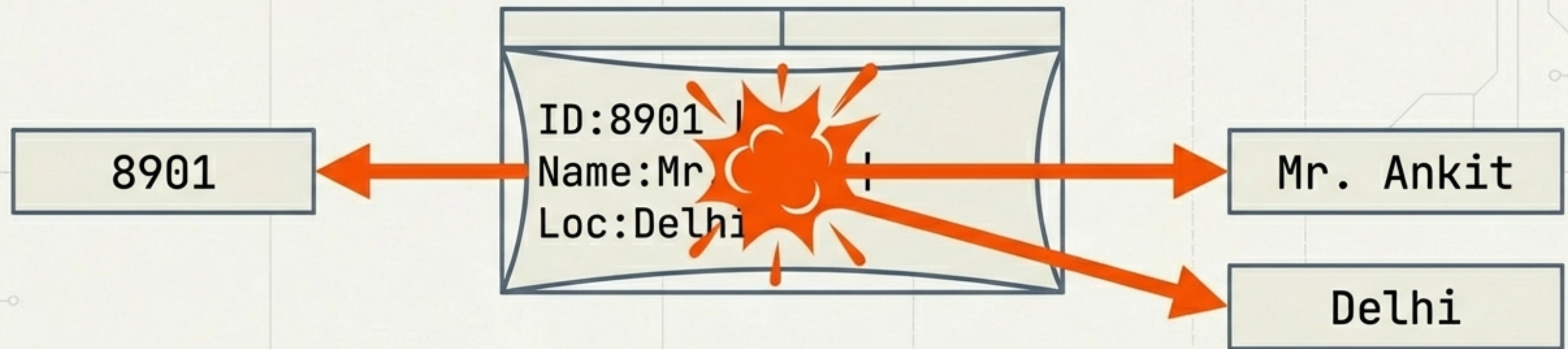
The Proof of Value.

1. We drop the noisy SibSp and Parch columns.
2. We retain only the engineered Family_Type.
3. The model gains a clearer signal about social structures.

While the accuracy lift here is modest, this technique is critical in larger datasets where feature interaction drives predictions.

Introduction to Feature Splitting

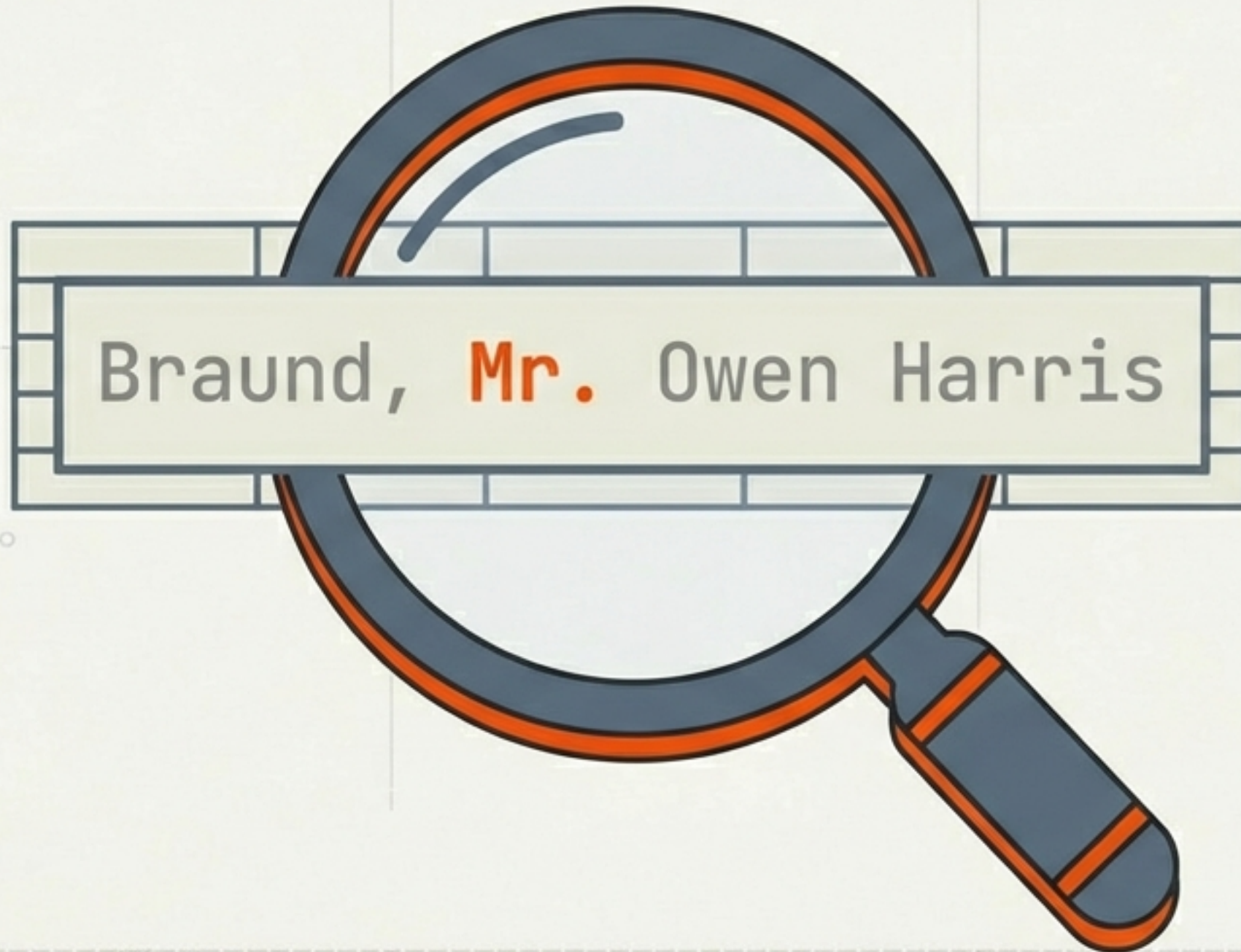
Solving the problem of Non-Atomic Data



Definition: Feature Splitting is the process of dissecting a single feature to extract multiple distinct pieces of information.

The Problem: 'Composite Data' prevents models from isolating variables. You cannot correlate 'Location' if it is trapped inside a string with a 'Name'.

Case Study II: The Hidden Value in Names

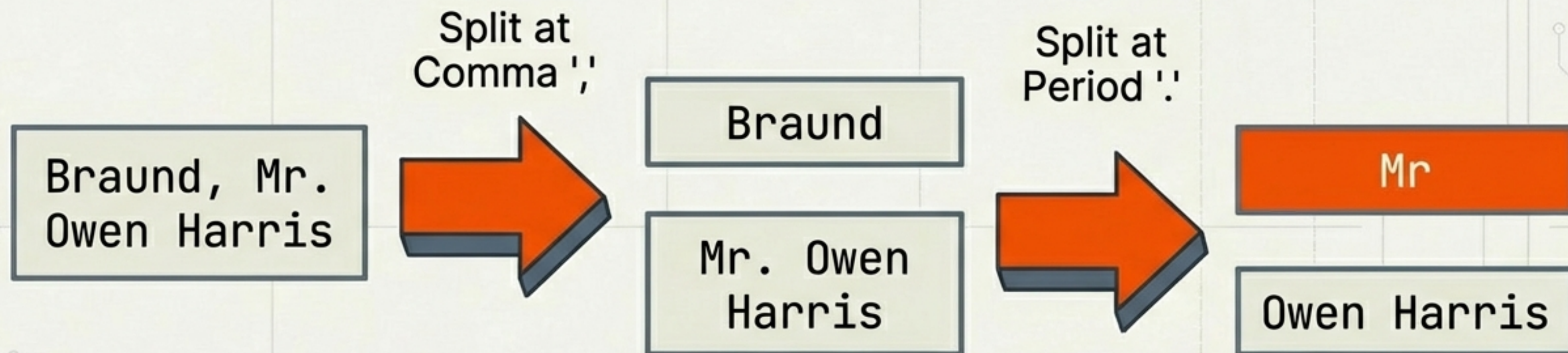


The 'Name' column appears mathematically useless because every entry is unique.

However, it contains a structural pattern: The Salutation (Title).

- The Title acts as a proxy for:
 1. Gender (Mr vs Mrs)
 2. Age (Master vs Mr)
 3. Marital Status (Miss vs Mrs)

The Splitting Mechanics



By parsing the string delimiters, we extract a clean categorical feature containing standardized values:

```
[ 'Mr', 'Mrs', 'Miss', 'Master', 'Dr', ... ]
```

Uncovering Hidden Correlations

Survival Rate by Title

Mrs
Married Women

79%

Miss
Single Women

70%

Master
Male Children

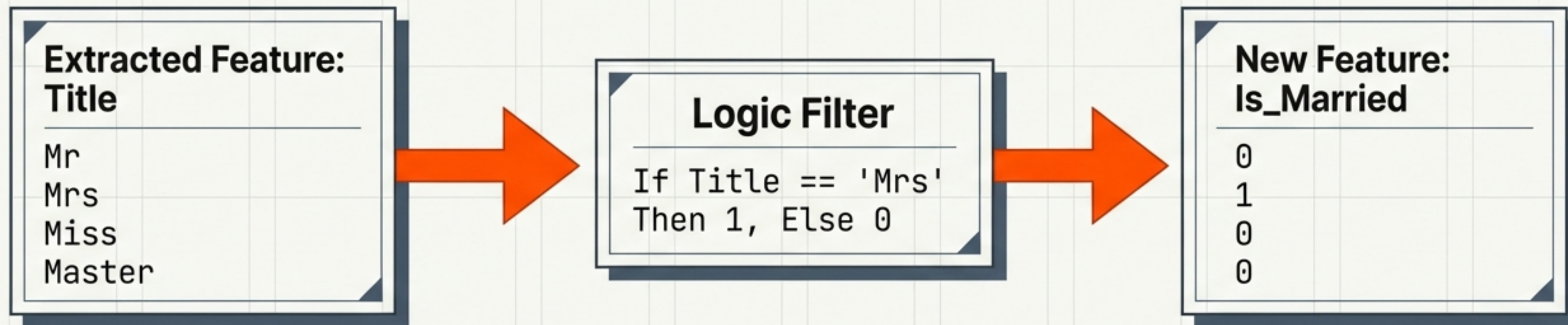
57%

Mr
Adult Men

15%

Insight: 'Master' refers to male children. The model can now distinguish between men (low survival) and boys (higher survival), a nuance lost in the raw 'Sex' column.

Advanced Iteration: Constructing “Is_Married”



Feature Engineering is recursive. We used Splitting to get the Title, and now we use Construction to derive Marital Status.

Finding: Married women (**Mrs**) had a **~79%** survival rate compared to **~70%** for unmarried women (**Miss**). This granular distinction creates a highly predictive signal.

Technique Comparison

Feature Construction



Logic

- Combining columns ($A + B = C$)

Use Case

- When the interaction between features matters more than the features themselves (e.g., Total Family Size).

Key Requirement

- Domain Intuition.

Feature Splitting



Logic

- Breaking columns ($A \rightarrow B, C$)

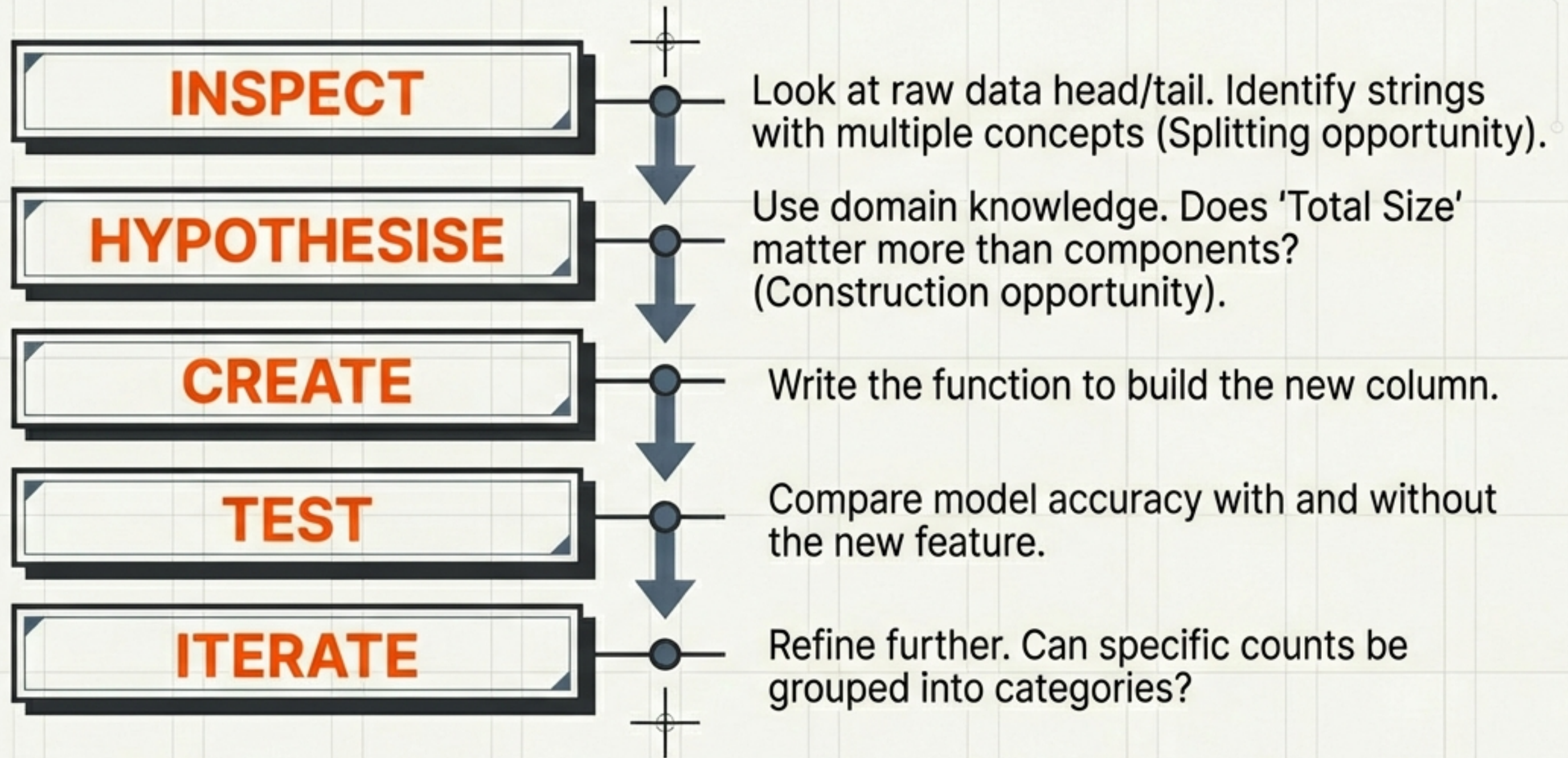
Use Case

- When a single cell holds multiple “non-atomic” data points (e.g., Names, messy date strings).

Key Requirement

- Parsing & Inspection.

Recommended Workflow



This recursive process drives continuous feature engineering improvement.

Practice & Application

Mastery requires practice.
These patterns exist in almost every dataset. Look for them in sports, finance, or logs.

Input Data

Runs Scored: 45
Balls Faced: 30



New Feature (Output)

Strike Rate
 $(45 / 30) * 100 = 150.0$



“The limit is not the mathematics, but your creativity. Always ask:
‘Can I create a new, more useful feature from what I already have?’”